Mathesh S

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EXECUTIVE SUMMARY

Skilled in CAD modeling, FEA, CFD, and multidisciplinary system design, I bring a strong simulation-driven approach to product engineering. The role requires bridging theoretical physics with functional, manufacturable designs, which aligns perfectly with my project experience in automotive, aerospace, and IoT-integrated systems. I am confident in my ability to translate concepts into real-world performance and eager to contribute effectively as a Design Engineer.

TECHNICAL SKILLS

- **CAD & 3D Modeling:** SolidWorks (Parametric, Tolerancing), Solid Edge, Fusion 360, Web-based 3D visualization (React, Three.js)
- **Simulation & Analysis:** ANSYS Fluent (CFD), ANSYS Static Structural (FEA), Kinematic & Dynamic Motion Analysis
- **Software & Programming:** Python, HTML, CSS, Machine Vision (YOLOv8)
- **Multidisciplinary Tech:** IoT Telemetry, Digital Twin modeling, Embedded Systems, Relay-Based Automation

SOFT SKILLS

- Problem Solving & Critical Thinking
- Cross-functional Teamwork
- Adaptability in Multidisciplinary Projects
- Technical Communication
- Time Management

PROFESSIONAL EXPERIENCE

Smart India Hackathon 2026 (AYASK) **Aug 2026 – Present**

Role: Multidisciplinary Project Lead

Responsibilities:

- Formed and led a multidisciplinary team to create a predictive conveyor belt monitoring system.
- Integrated PatchCore and YOLOv8 machine vision models for real-time defect detection.
- Developed IoT telemetry frameworks and digital twin models for remote structural monitoring.

MSME Idea Hackathon 6.0 (Omni-Node) **2026**

Role: Hardware Architecture Contributor

Responsibilities:

- Developed an AI-Driven Retrofit Ecosystem aimed at High-Mix Low-Volume (HMLV) manufacturing flexibility.
- Drafted detailed hardware component architectures and comprehensive system block diagrams.
- Created visual presentation frameworks for pitching the hardware-software integration concept to stakeholders.

Impact: Formulated a viable retrofit framework bridging modern AI software with legacy manufacturing hardware.

Dassault Systèmes AAKRUTI Initial Design Challenge **2026**

Role: Design Competitor

Responsibilities:

- Designed a community greywater recycling project concept named "AquaHarvest".
- Created comprehensive system documentation detailing fluid flow routing and mechanical integration.
- Generated high-quality visual rendering submission slides representing the finalized product layout.

Impact: Delivered a sustainable, conceptually validated engineering solution for urban water reuse and conservation.

4-Cylinder Engine Assembly **2025 – 2026**

Role: Mechanical Design Contributor

Responsibilities:

- Designed a full parametric 3D assembly of a 4-cylinder internal combustion engine using SolidWorks.
- Modeled all major components including pistons, crankshaft, connecting rods, cylinder block, and valvetrain.
- Engineered strict tolerance stacks and ran interference checks across mating components.
- Simulated dynamic motion of the piston-crankshaft mechanism under simulated operating loads.

Impact: Validated kinematic behavior and ensured 100% assembly feasibility and mechanical fit before physical prototyping.

Turbulent Flow Over a Backward-Facing Step **2025 – 2026**

Role: CFD Analyst

Offline Voice-Controlled Switch **2025**

Role: Electronics Integration Lead

Responsibilities:

- Designed a localized voice-controlled switch using the Ai-Thinker VC-02 offline speech recognition module.
- Integrated a 4-channel relay module to independently actuate multiple electrical components via voice commands.
- Eliminated dependency on cloud-based voice processing and internet connectivity.

Impact: Built a fully functional, zero-latency automation prototype capable of reliable offline physical actuation.

EDUCATION

Bachelor of Engineering (B.E.), Mechanical Engineering **Aug 2025 – Present**

Sri Krishna College of Engineering and Technology (SKCET), Coimbatore

AWARDS & ACCOMPLISHMENTS

- Participating in Dassault Systèmes AAKRUTI Design Challenge (2026).
- Competitor & Team Lead in Smart India Hackathon 2026 (AYASK & Tarang).
- Competitor in MSME Idea Hackathon 6.0 (Omni-Node).
- Currently preparing for the Certified SOLIDWORKS Professional (CSWP) examination.